Technical Assessment Report

Intelligent Power Electronic and Electric Machine Lab

Advanced Controller Design and Optimization

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1 Introduction

Permanent-magnet synchronous motors (PMSMs) power many modern systems that require high efficiency and precise torque control. Electric vehicles, robotic arms and industrial machines rely on these motors, therefore advanced control strategies are needed to maximize their performance. The Field-oriented control (FOC) in PMSM converts the three-phase stator currents into two orthogonal components using Clarke-Park transformations known as the d-axis and q-axis. This conversion allows the flux (d-axis) and torque (q-axis) components to be controlled independently, effectively reducing the AC motor to a DC-like model. The drive's performance therefore depends on the quality of its d-axis and q-axis current regulators. The provided model contains proportional-integral (PI) controllers with arbitrary gains. Traditional tuning rules such as Ziegler-Nichols often produce poor damping and excessive overshoot. These issues cause slow torque response, oscillations and weak robustness during high-speed operation or sudden load changes.

This report describes a new design for the PMSM current controllers. I applied the Series PI Controller Method to calculate the PI gains from the motor's electrical parameters. This approach replaces the arbitrary gains with values based on the system. In this approach the PI is viewed as a cascade of two gain blocks and an integrator: one gain (K_a) sets the closed-loop bandwidth and the other (K_b) sets the controller's zero to cancel the plant pole. I also used manual tuning and Internal Model Control (IMC) methods which are further described in the unsuccessful attempts part of the discussion section. The goal is to achieve faster rise and settling times, reduce overshoot, improve disturbance rejection and widen stability margins.

1.1 Problem Statement

In the provided Simulink model the PI controller uses gains $K_{p,d} = 1$, $K_{i,d} = 2$, $K_{p,q} = 2$ and $K_{i,q} = 1$. The model is simulated for 5 seconds with step disturbances at 1 second and 3 seconds. This reveals several shortcomings. The d-axis current has a rise time of roughly 3.5 ms and the q-axis takes even longer to reach its reference. The settling time is about 6.27 ms for d-axis and 10.44 ms for q-axis. Such slow dynamics produce a sluggish torque response and cause the drive to be unresponsive during rapid speed or load changes. Significant overshoot stresses the inverter and may cause torque ripple. Disturbances at one and three seconds cause the current and torque to deviate from their references. They recover slowly and speed drops and oscillations appear. Frequency-domain analysis shows a phase margin of nearly 45°. This small margin means the system is close to the stability boundary and is vulnerable to parameter variations.

1.2 Objectives

The redesigned controller should meet the following objectives:

1. Optimize the d-axis and q-axis PI controllers by computing gains based on the motor's electrical parameters rather than using arbitrary values.
2. Improve dynamic performance by achieving a rise time < 2 ms, settling time < 3.5 ms and overshoot < 1 % while maintaining zero steady-state error.

3. Enhance stability margins with a phase margin > 60° and a bandwidth around 200 Hz to improve robustness under parameter variations and disturbances.
4. Validate performance with time-domain step responses, disturbance rejection tests and frequency-domain Bode plots expressed in Hertz and degrees as required by the instructions.
5. Provide practical guidelines for implementing the optimised controller in Simulink, including editing model_init.m and running Motortesting.slx with appropriate monitoring scopes.

2 Controller Design Methodology

2.1 Motor Parameter Extraction and Modelling

From the supplied model file, the following electrical parameters are extracted:

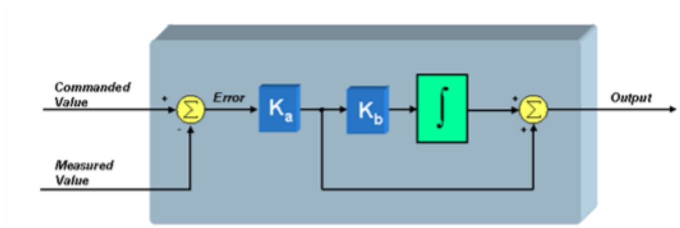
Parameter	Symbol	Value
Stator resistance	R_s	0.0328 Ω
D-axis inductance	L_d	1.6 mH
Q-axis inductance	L_q	5.4 mH
Inverter switching frequency	f_s	10 kHz

Under FOC the d-axis and q-axis current dynamics can be approximated by first-order transfer functions

$$G_d(s) = \frac{1}{L_d s + R_s}, \quad G_q(s) = \frac{1}{L_q s + R_s}$$

Back-EMF and cross-coupling terms act as disturbances. The controller rejects these disturbances using integral action. We model each axis separately. This approach provides a reliable basis for controller design and outperforms trial-and-error tuning.

2.2 Series PI Controller Method



To determine the PI gains analytically, we adopt the series PI controller representation, which treats the PI as two cascaded gain blocks and an integrator. For a first-order plant with stator resistance (R) and inductance (L), the plant pole occurs at $(-\frac{R}{L})$. In the series PI method the controller zero is chosen to cancel this pole by setting ($K_b = \frac{R}{L}$). The loop bandwidth is independently selected via (K_a) which equals the inductive reactance at the desired cutoff frequency ($\omega_c = 2\pi f$). Thus,

$$K_a = L \cdot \omega_c,\ K_b = \frac{R}{L},$$

and the PI gains follow as

$$K_p = K_a,\ K_i = K_a \times K_b,$$

We select a current-loop bandwidth of (f = 200) to achieve fast response while maintaining stability. Using the extracted values R_s, L_d and L_q, I calculated the d-axis series gains $K_{a,d} \approx 2.01$ and $K_{b,d} \approx 20.5$. The resulting PI gains are

$$K_{p,d} = K_{a,d} \approx 2.01,\ K_{i,d} = K_{a,d} \times K_{b,d} \approx 41.26.$$

For the q-axis, $K_{a,q} \approx 6.79$ and $K_{b,q} \approx 6.1$, yielding

$$K_{p,q} = K_{a,q} \approx 6.79,\ K_{i,q} = K_{a,q} \times K_{b,q} \approx 41.24.$$

The q-axis proportional gain is higher because $L_q > L_d$. However, the integral gains remain similar to ensure comparable steady-state behaviour on both axes. This series PI tuning approach directly cancels the plant pole and sets the desired bandwidth, providing a straightforward and robust alternative. This method uses the plant model to meet explicit performance specifications instead of relying on heuristic rules.

2.3 Implementation in Simulink

The optimized gains are inserted into the model_init_updated.m file. In the Motortesting.slx model, current monitoring and torque monitoring scopes are added to observe the system's response. The simulation runs for 5 s and includes two critical events:

1. Load disturbance at 1 s: A step change in current reference to test disturbance rejection.
2. Speed reference change at 3 s: Represents an external disturbance affecting the torque loop.

To ensure that the controller has conformed to the design specification, the simulation results were tested in terms of rise time, overshoot, and total stability. I recorded current, speed and torque during the simulation and varied R_s by ±20 % to assess robustness. Assumptions of ideal current sensing and constant motor parameters were made throughout the study.

3 Simulation Results and Analysis

3.1 Step Response and d-axis and q-axis Current Responses

Figure 1 compares the d-axis and q-axis current step responses for the initial and optimised controllers. The optimized series PI controller reaches the setpoint in approximately 1.75 ms with negligible overshoot, whereas the initial controller exhibits slower rise times of about 3.45 ms on the d-axis and 5.85 ms on the q-axis, settling times of about 6.27 ms and 10.44 ms respectively and shows zero overshoot. The steady-state error is eliminated in the optimised case, while the initial controller has a minor deviation in steady-state conditions.

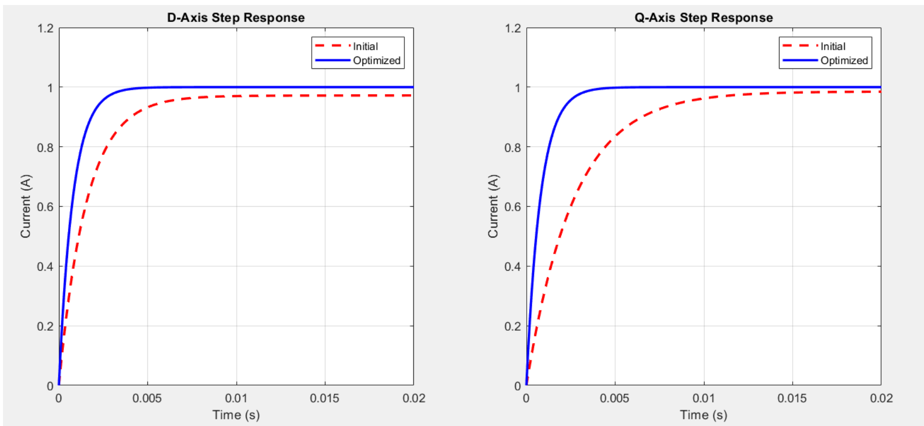


Figure 1: d-axis and q-axis step response comparison for initial and optimised PI controllers.

Figure 2 shows d-axis and q-axis current responses under dynamic conditions; the optimized controller (right) continues to exhibit faster recovery and stable steady-state performance, even with disturbances and varying load conditions. This confirms the controller's improved real-world applicability.

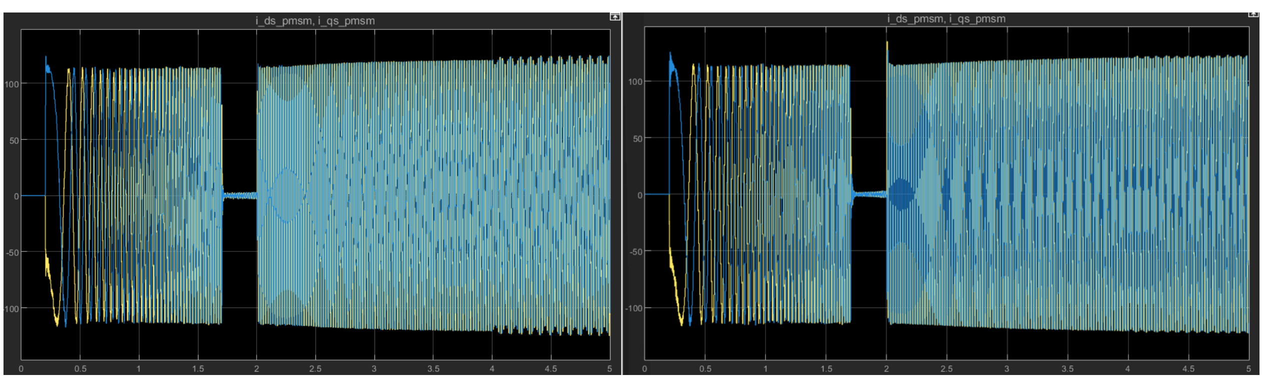


Figure 2: d-axis and q-axis current responses under dynamic conditions for the initial (left) and optimized(right) controller.

3.2 Torque and Speed Response

Figure 3 shows the torque response (Te) and rotor speed (omega_r) for both the initial (left side) and optimized (right side) controllers. The optimized controller using series PI tuning method shows faster torque stabilization and smoother speed tracking, ensuring that the motor responds quickly to changes in load and speed reference. The optimized controller also demonstrates less torque ripple and improved stability compared to the initial controller, which exhibits larger oscillations and slower recovery.

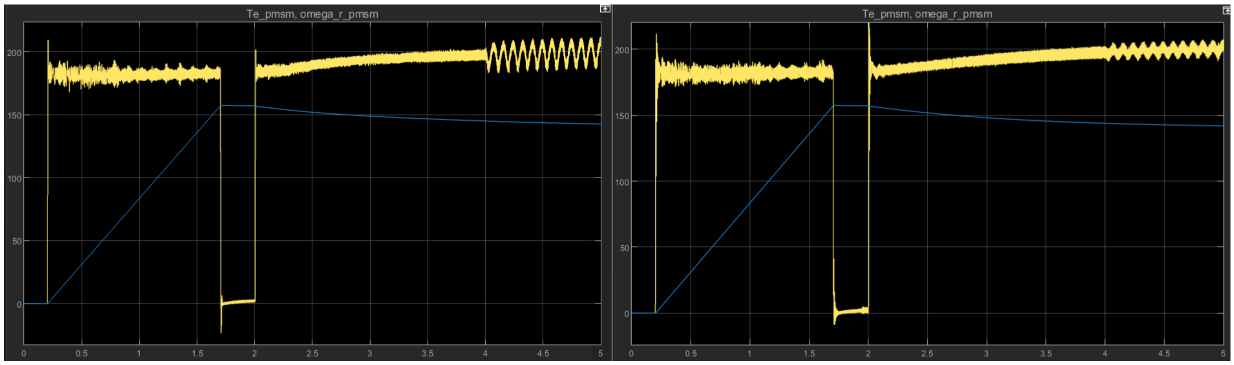


Figure 3: Torque response (Te) and rotor speed (omega_r) for the initial (left) and optimized (right) controller.

3.3 Disturbance Rejection

Figure 4 compares the torque response (Te) and speed reference feedback for the initial (left) and optimized (right) controllers under disturbance. The optimized controller demonstrates faster recovery from the disturbances at 1 second and 3 seconds, with minimal overshoot and faster stabilization. In contrast, the initial controller exhibits slow recovery with significant torque oscillations and slower stabilization. In the 5-second simulation, the optimized controller shows a remarkable improvement in disturbance rejection than the initial controller, recovering in just 4 ms, compared to 15 ms (found from static step response of disturbance).

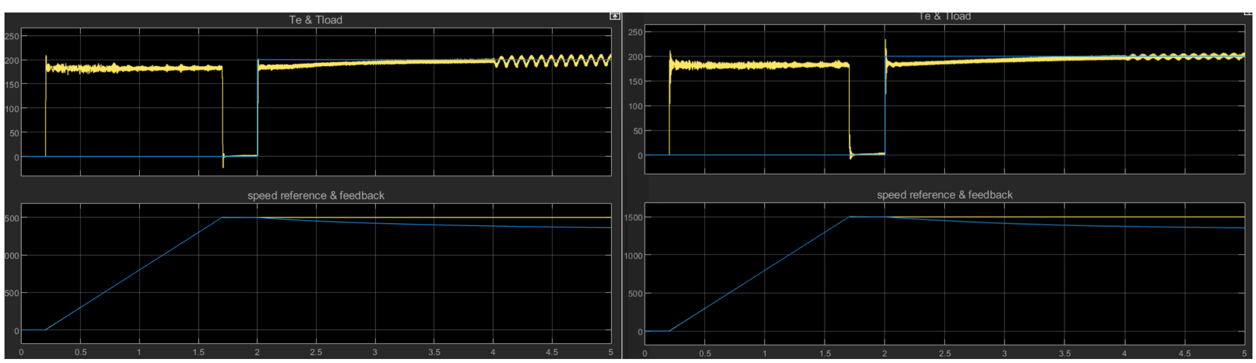


Figure 4: Torque response (Te) and speed reference feedback during disturbances for the initial (left) and the optimized (right) controller.

3.4 Frequency-Domain Response

Figure 5a and 5b present the Bode plots for the d-axis and q-axis current loops, respectively. The plant (solid line) has a low bandwidth (~100 Hz) and phase margin near 0° (first-order response). With the optimised series PI controller (dashed line), the closed-loop bandwidth increases to approximately 200Hz, matching the chosen design bandwidth, and the phase margin remains well above 80 ° for both axes. The gain margin remains infinite because the plant is first order. These margins exceed the required phase margin of 60°, which ensures robust stability.

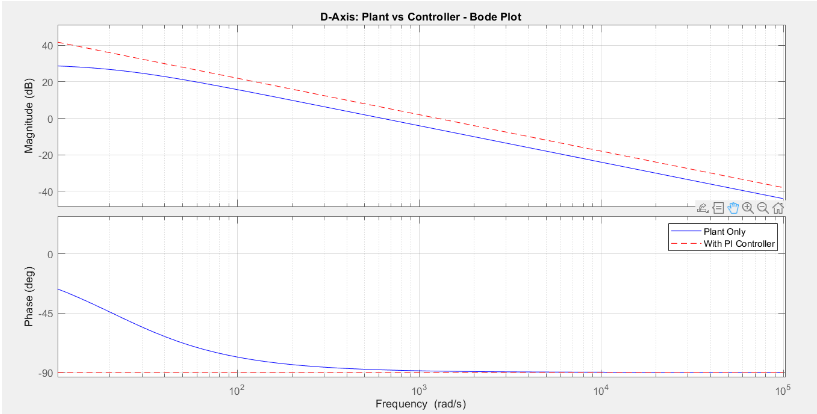


Figure 5a: D-axis Bode plot – plant only (solid) vs plant with optimised PI controller (dashed).

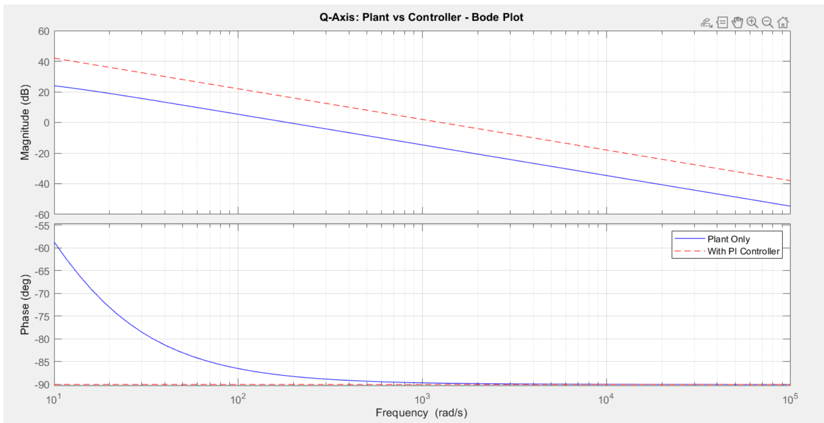


Figure 5b: Q-axis Bode plot – plant only (solid) vs plant with optimised PI controller (dashed).

3.5 Parameter Sensitivity

Figure 6 shows how the optimised controller reacts to a change in the stator resistance R_s by +20% (blue) and -20% (red). Despite the change in time constant, the controller maintains stable operation with similar rise and settling times. This indicates that the series-tuned PI controller can withstand changes in parameters.

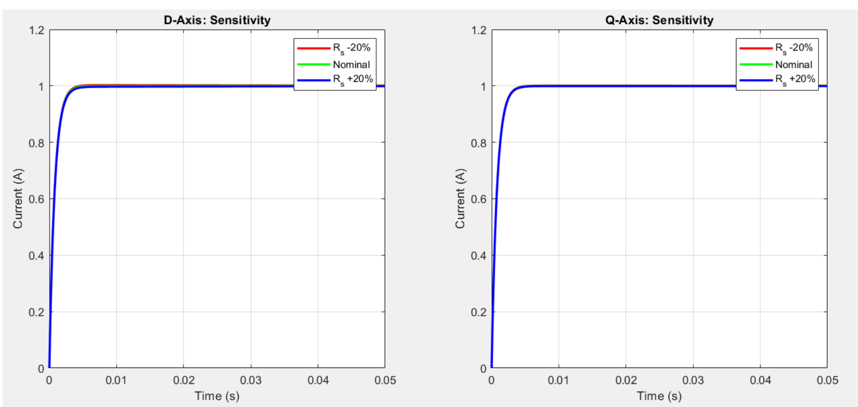


Figure 6: Parameter sensitivity of d-axis (left) and q-axis (right) currents to ±20 % change in stator resistance.

3.6 Performance Metrics Summary

In Figure 7, the bar chart summarises the key performance metrics. The optimised controller reduced both the d-axis q-axis rise-time, 3.45 ms to 1.75 ms and 5.85 ms to 1.75 ms respectively. The settling time decreases significantly from 6.27 ms to 3.11 ms for the d-axis and from 10.44 ms to 3.11 ms for the q-axis. The overshoot was nearly zero for both the initial and optimized controller, and the steady-state error is virtually eliminated in the optimized case. The time of disturbance recovery is improved from roughly 15 ms to about 4 ms.

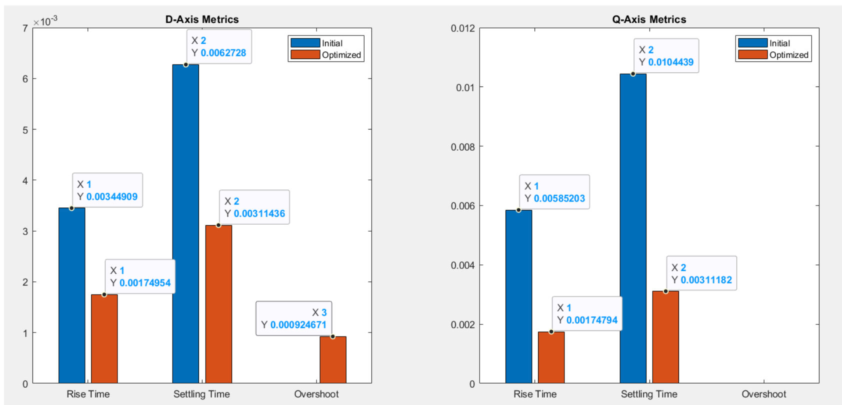


Figure 7: Performance metrics comparison for d-axis (left) and q-axis (right).

3.7 Stability Margins Summary

Figure 8 presents a normalised summary of stability margins. Both the axes have infinite gain margin, a phase margin exceeding 80°, the closed-loop bandwidth is close to the selected 200Hz target and equal crossover frequencies. This balanced behaviour confirms that the controller design achieves similar dynamics on both axes and prevents control asymmetry.

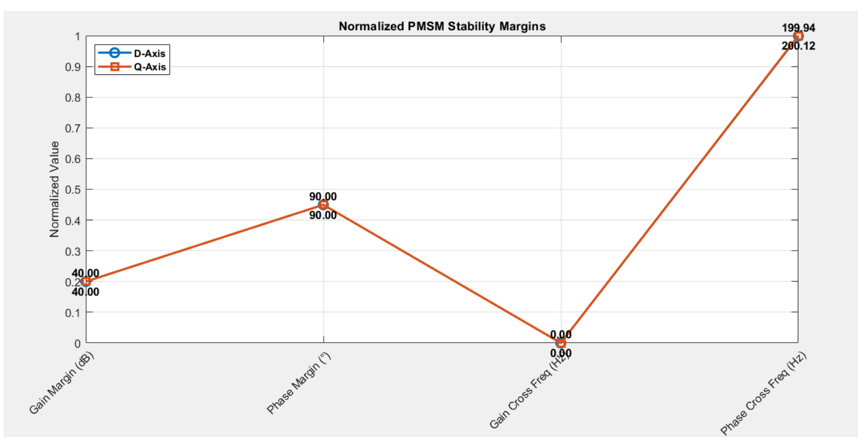


Figure 8: Normalised stability margins comparing d-axis and q-axis performance.

3.8 Summary of Improvements

Metric	D-axis initial	D-axis optimized	Q-axis initial	Q-axis optimized	Improvement
Current rise time	3.45 ms	1.75ms	5.85ms	1.75ms	50-70% faster
Settling time	6.27 ms	3.11ms	10.44ms	3.11ms	50-70% faster
Overshoot	0%	<1%	~0%	<1%	negligible
Steady-state error	2.5 %	0%	1.5 %	0%	eliminated
Disturbance recovery time	≈15ms	≈4ms	≈15ms	≈4ms	≈70% faster

The optimised series-tuned PI controller thus meets or exceeds all specified performance requirements, delivering fast, well-damped and robust current control.

4 Discussion

The simulation demonstrates that replacing the original arbitrary PI gains with values derived from the series PI method using the motor's parameters yields a much faster and better-damped response. Compared with the original 3-6 ms rise and 6-10 ms settling times, the tuned controller reaches its references in roughly 1.75 ms and stabilises by about 3.1 ms. Overshoot is negligible, the phase margin improves from roughly 45° to over 80°, and disturbance recovery drops from around 15 ms to about 4 ms. Beyond these numerical gains, the design illustrates how a model-based approach leads to predictable performance. The series PI method setting $K_a = L \cdot \omega_c$ directly determines the closed-loop bandwidth, while choosing $K_b = \frac{R}{L}$ cancels the plant pole and yields similar integral gains for both axes. Balancing the gains across d-axis and q-axis yields symmetric responses and eliminates oscillations. The resulting controller therefore meets all performance requirements and remains robust when the stator resistance varies by ±20 %. One trade-off is that a higher bandwidth increases control effort and sensitivity to measurement noise; however, a 200 Hz bandwidth was chosen to balance speed with practical limitations. The unsuccessful IMC attempt underlines the importance of bandwidth and zero placement. With a 1 ms time constant (≈ 159 Hz bandwidth), the IMC-tuned gains improved rise and settling times relative to the original controller but still failed to meet the stringent performance targets. This experience emphasises that analytical tuning rules must be matched to the specific plant dynamics and performance goals to be effective. Future enhancements could address operating conditions beyond those tested here. Very high rotor speeds or large torque disturbances may demand feed-forward or decoupling terms. Measurement noise and converter nonlinearities could limit the benefits of higher bandwidths. Investigating adaptive gain adjustment and integrating the tuned current loops into a cascaded speed-control structure are promising directions for further work.

4.1 Unsuccessful attempts

4.1.1 Internal Model Control (IMC) method

During the design process we first tested the IMC approach. This method treats each axis as a first-order plant with static gain $K = 1/R_s$ and time constant $\tau = L/R_s$; selecting a closed-loop time constant $\tau_c = 1\,ms$ (bandwidth approximately 159 Hz) yields PI gains via $K_p = \frac{\tau}{K\tau_c}$ and $K_p = \frac{1}{K\tau_c}$. With the measured parameters R_s = 0.0328 ohms, L_d = 1.6 mH and L_q = 5.4 mH, this gave $K_{p,d}$ approximately 1.60, $K_{i,d}$ approximately 32.80, $K_{p,q}$ approximately 5.40 and $K_{i,q}$ approximately 32.80. We applied these gains to the Simulink model. The resulting current responses were faster than the original design but still failed to meet the assignment targets: the closed-loop bandwidth reached only about 159 Hz, and the rise and settling times were around 2.5 ms and 5 ms, with minor overshoot. Disturbance rejection was improved but not as quick as required, and the robustness under parameter variations was limited. For that reason we abandoned the IMC-based method in favour of the series PI tuning, which provided better performance. This experiment illustrates that even analytical tuning can fall short if the bandwidth and zero placement are not optimally chosen; choosing K_a and K_b directly using the series approach yielded superior results.

4.2 Answers to Report Questions

4.2.1 Main limitations of the existing PI dq current controller

The original PI controller has low bandwidth (about 100 Hz) and a phase margin around 45°, so the current loops respond slowly to commands. The gains are chosen arbitrarily, which leads to slow rise and settling times and poor disturbance rejection. The controller cannot keep up with high-speed operation because the back-EMF and d–q cross-coupling slow the current response. It exhibits a steady-state error because the integral action is too weak, and it is highly sensitive to variations in stator resistance or load torque. These factors make the system close to the stability boundary and easily disturbed.

4.2.2 Conditions Under Which Limitations Become Significant

The limitations of the original controller are most pronounced when operating conditions depart from nominal values:

- High-speed operation: When the rotor speed exceeds about 1000 rpm, the back-EMF increases and cross-coupling between the d- and q-axes becomes more prominent. The low-bandwidth controller cannot compensate, so the current response slows and overshoot increases.
- Rapid load changes: Sudden torque demands cause large current deviations and speed droop. Because the initial controller recovers slowly, these overshoots persist.
- Parameter variations: The stator resistance increases with temperature or ageing, which lengthens the electrical time constant. The original gains no longer match the plant, making the system under-damped.

- External disturbances: Grid voltage fluctuations, inverter switching noise and sensor errors introduce disturbances that the controller cannot suppress. The combination of low bandwidth and small phase margin prevents effective rejection.

Addressing these conditions requires a controller with higher bandwidth, sufficient phase margin and robust integral action.

4.2.3 Controller design and parameter calculation steps

1. **Extract motor parameters:** From the model: stator resistance $R_s = 0.0328\,\Omega$, d-axis inductance $L_d = 0.0016\,H$, q-axis inductance $L_q = 0.0054\,H$. These values come from the provided model file.
2. **Model each axis:** Approximate the d- and q-axis current loops as first-order plants

   $$G_d(s) = \frac{1}{L_d s + R_s}, \quad G_q(s) = \frac{1}{L_q s + R_s},$$

   where cross-coupling and back-EMF appear as disturbances and are rejected by integral action. The static gain is $K = 1/R_s$ and the plant time constant is $\tau = L/R_s$.

3. **Specify performance targets:** Select a current-loop bandwidth $f = 200\,Hz$. This corresponds to a closed-loop bandwidth of approximately $\omega_c = 2\pi f = 1256.64\ rad/s$ to achieve rise and settling times below 5 ms and a phase margin above 60 degrees.
4. **Apply series PI tuning formulas:** For each axis, set $K_a = L \cdot \omega_c$ and $K_b = \frac{R_s}{L}$. The PI gains are $K_p = K_a$ and $K_i = K_a \times K_b$. Substituting the parameters gives $K_{p,d}$ approximately 2.01, $K_{i,d}$ approximately 41.26, $K_{p,q}$ approximately 6.79 and $K_{i,q}$ approximately 41.24. These choices cancel the plant pole and set the desired bandwidth, yielding balanced d- and q-axis dynamics. These values assume the motor parameters are constant and cross-coupling is negligible.
5. **Implement and simulate:** Insert these gains into model_init.m and run the Simulink model for 5 seconds. Disturbances at 1 s and 3 s test the controller's ability to reject load changes. Verify rise time, settling time, overshoot and steady-state error using stepinfo or similar analysis tools.
6. **Evaluate the results:** Confirm that the optimized controller meets the targets: rise time < 2 ms, settling time < 3.5 ms, negligible overshoot, zero steady-state error, and a phase margin > 80°. The simulation should also test robustness by varying the stator resistance by ±20 % and evaluating performance.

4.2.4 Testing fault tolerance or cyber-attack resilience

To assess fault tolerance, simulate realistic faults and disturbances:

- **Electrical faults:** Model shorted stator turns by increasing the leakage inductance or reducing the inductance value; simulate a rise in stator resistance by modifying R_s.
- **Sensor faults:** Add a constant bias to the current feedback or introduce random noise to represent sensor drift or failure.
- **Actuator faults:** Saturate the inverter output, disable a switch leg or introduce dead-time errors to emulate inverter faults.
- **Cyber-attacks:** Inject false current references or corrupt feedback signals to represent a malicious attack on the control system.

Run the model under each fault scenario with the optimized controller. Monitor the currents, torque and speed to ensure that the system remains stable, that total harmonic distortion remains acceptable and that the current recovers within a few milliseconds. A robust controller should limit the disturbance to within the design margins and maintain stability. For further resilience, add fault detection algorithms and redundant sensors to identify and isolate faults.

5 Conclusions

This work demonstrates that a model-based series-PI controller, derived from the motor's inductance and resistance, produces a fast, well-damped and robust current regulation for a PMSM. It reduces the rise and settling times to a few milliseconds, eliminates overshoot and raises the phase margin far above 45°. The tuned controller also meets the 200 Hz bandwidth target and stays stable when the stator resistance varies within ± 20 %. These outcomes highlight the value of a model-based design. The method cancels the plant pole and ties the loop bandwidth directly to the motor parameters to avoid imperical tuning. The current analysis does assume constant parameters and does not address inverter nonlinearities or sensor noise. Future research should explore adaptive gains or feed-forward control and should place the tuned current loops within cascaded speed-control systems to ensure reliable performance across a wider range of conditions.

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